

2(a)	$F = \rho g V$	C1
	$0.071 = 1.2 \times 9.81 \times V$	
	$(V = 6.03 \times 10^{-3} \text{ m}^3)$ $\frac{4}{3} \times \pi \times r^3 = 6.03 \times 10^{-3}$ $r = 0.11 \text{ m}$	A1
2(b)	$m = 0.053 / 9.81$	C1
	$(= 5.4 \times 10^{-3} \text{ kg})$	
	$F = 0.071 - 0.053$ $(= 0.018 \text{ N})$	C1
	$a = (0.071 - 0.053) / (0.053 / 9.81)$ $= 3.3 \text{ m s}^{-2}$	A1
2(c)(i)	$v^2 = u^2 + 2as$	C1
	$3.6^2 = (-1.4)^2 + 2 \times 9.81 \times s$ or $3.6^2 = 1.4^2 + 2 \times 9.81 \times s$	
	$s = 0.56 \text{ m}$	A1
2(c)(ii)	single straight line from any positive non-zero value of v at $t = 0$ to any negative non-zero value of v at $t = T$	B1
	line starting at $(0, 1.4)$ and ending at $(T, -3.6)$	B1